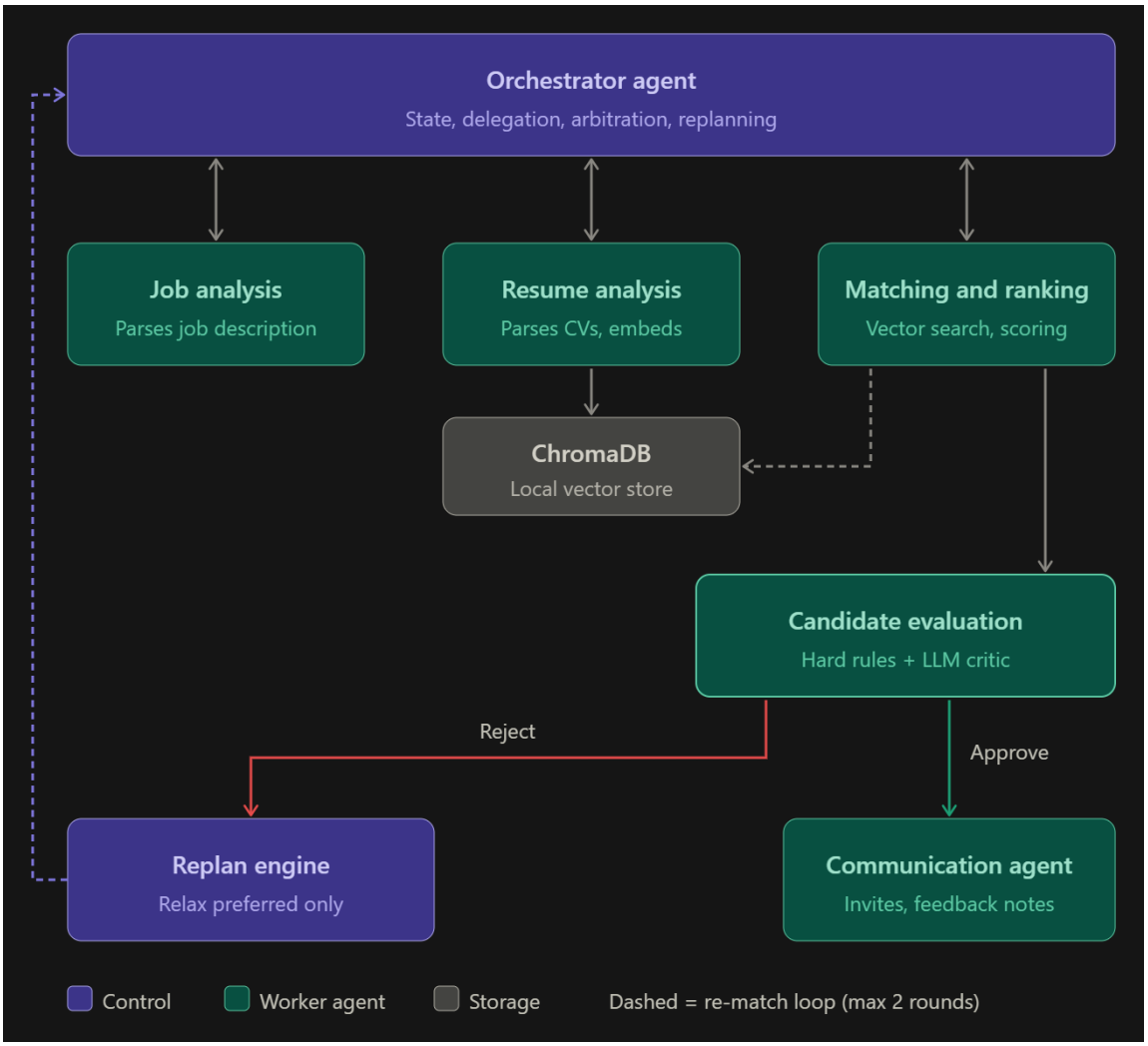
TalentAI: Multi-Agent Intelligent Recruitment System

Technical Design Document -- Architecture, Roles, Coordination & Tech Choices

1. System Architecture & Communication Topology

TalentAI implements a **Centralized Orchestrator-Worker topology** with an independent quality evaluation gate. Inter-agent communication flows deterministically through the central Orchestrator, ensuring state consistency and bounded execution.



2. Agent Roles and Responsibilities

Each agent executes a defined single-responsibility role with structured Pydantic inputs and outputs:

Agent	Role & Responsibilities	Input / Output	Tools & Tech
1. Orchestrator Agent	Central coordinator. Dispatches tasks across agents, maintains canonical state, detects Evaluation conflicts, and triggers dynamic replanning with bounded iterations (max 2).	**In:** RecruitmentRequest **Out:** RecruitmentState, SSE Stream	CrewAI Agent, State Machine, Groq LLaMA-3.3-70B
2. Job Analysis Agent	Parses raw job postings into non-negotiable mandatory skills, preferred qualifications, experience bounds, role title, and company profile.	**In:** Raw JD string **Out:** JobDescription Model	Groq LLaMA-3.3-70B (JSON entity extraction)
3. Resume Analysis Agent	Parses candidate resumes in parallel. Extracts verified technical skills, professional vs. internship experience, and indexes dense vectors into ChromaDB.	**In:** Raw Resumes list **Out:** Candidate Model list	Groq LLaMA-3.1-8B-Instant, ChromaDB Ingestion
4. Matching & Ranking Agent	Executes semantic vector search across candidate embeddings and applies composite scoring formula (Mandatory 60%, Preferred 30%, Experience 10%). Supports relaxed rerun mode.	**In:** JobDescription, Candidates **Out:** MatchedCandidate list	RAGRetrievalTool (ChromaDB), ScoringTool
5. Candidate Evaluation Agent	Independent quality gate. Implements a Dual-Stage Hybrid Critic: Stage 1 deterministically checks 100% mandatory compliance; Stage 2 performs qualitative audit. Emits APPROVE or REJECT.	**In:** Shortlist, JobDescription **Out:** Verdict, Audit Notes	Independent LLM Call, Deterministic Rule Engine
6. Communication Agent	Generates personalized, professional interview invitations highlighting candidate achievements, and respectful, constructive feedback notices for non-selected applicants.	**In:** Evaluated Shortlist **Out:** Formatted Outreach Texts	Groq LLaMA-3.3-70B (Generative NLG)

3. Orchestration Pattern & Conflict Resolution Protocol

Pattern: Centralized Orchestrator-Worker with Adversarial Feedback Loop

A centralized Orchestrator-Worker pattern was selected over peer-to-peer (P2P) for three foundational reasons: (1) *Single Source of Truth:* Centralized state ensures candidate records remain synchronized; (2) *Deadlock Elimination:* Unbounded negotiation cycles between agents are strictly bounded by MAX_REPLAN_ROUNDS = 2; (3) *Audit Compliance:* All decisions and conflict logs are serialized sequentially for recruiter review.

Conflict Resolution Protocol:

- **Adversarial Quality Gate:** If a candidate (e.g. Arjun Mehta) achieves high vector similarity but lacks mandatory skills (missing TypeScript and having only 3 years experience against a 4+ year requirement), the Evaluation Agent issues an explicit **REJECT** conflict signal.
- **Orchestrator Strategic Decision:** The Orchestrator receives the rejection and executes orchestrator_decide_replan() to select from: replan_relaxed, replan_strict, escalate, or accept_partial.
- **Strict Mandatory Preservation Rule:** Under replan_relaxed, preferred criteria (AWS, GraphQL) are relaxed, but **mandatory criteria remain 100% non-negotiable**.
- **Bounded Execution:** Maximum 2 replan rounds prevent infinite looping.

4. Technology Choices & Architectural Justifications

Each framework and library was selected to optimize latency, deterministic reliability, and local zero-cost execution:

Component	Technology Selected	Version / Spec	Engineering Justification
Agent Framework	CrewAI	1.15.x	Native Agent, Task, and Crew abstractions; clean tool bindings; supports structured Pydantic schemas and deterministic multi-agent workflows.
Primary LLM	Groq LLaMA-3.3-70B	70B Versatile	Sub-second inference (<800ms) on Groq LPUs prevents timeouts during multi-turn orchestration. High reasoning quality for JD extraction and qualitative auditing.
Fast Parser LLM	Groq LLaMA-3.1-8B	8B Instant	High-throughput, cost-effective model for bulk resume entity extraction without sacrificing JSON structured accuracy.
Vector Database	ChromaDB	0.5.x / 1.x	In-process vector store with zero cloud latency, zero external API costs, local persistence, and native cosine similarity retrieval.
Embedding Model	sentence-transformers	all-MiniLM-L6-v2 (384-d)	Lightweight local embeddings capturing semantic equivalence between resume phrasing and job specifications.
Backend API	FastAPI	0.115.x	Asynchronous event loop, native Server-Sent Events (SSE) streaming for real-time agent telemetry, and native Pydantic schema validation.
Frontend UI	Next.js 14	App Router / TS	Modern reactive interface, real-time EventSource telemetry streaming, interactive candidate review studio, and side-by-side match breakdown.